

Statistics 537, Fall 2026

Mixed Model Methods

Instructor: Gilbert W. Fellingham, Ph.D.

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Class Meeting Time 12:30– 1:45 TTh

Office Hours 10-11 T and 2-3 Th, or by appointment

A. Introduction

At the end of this course, we expect that students will be able to:

- i. explain the difference between fixed and random effects as implemented in the mixed model.
- ii. appropriately apply mixed models to deal with nonindependent data and a variety of covariance structures.
- iii. evaluate the fit of the mixed model using Choleski residuals.
- iv. perform frequentist analyses of mixed models using R and SAS.
- v. perform Bayesian analyses of mixed models using Nimble and/or Stan.

B. Required Text

None

C. Other sources of material

SAS® System for Mixed Models by Littell, Milliken, Stroup, and Wolfinger.

Analysis of Repeated Measures, Crowder and Hand

Analysis of Longitudinal Data, Diggle, Liang and Zeger

Linear Mixed Models in Practice, Verbeke and Molenberghs (Editors)

Mixed-Effects Models in S and S-Plus, Pinheiro and Bates

Applied Mixed Models in Medicine, Brown and Prescott

Linear Mixed Models for Longitudinal Data, Verbeke and Molenberghs

Hierarchical Linear Models, Bryk and Raudenbush

Models for Repeated Measurements, Lindsey

Random Coefficient Models, Longford

Statistical Tests for Mixed Linear Models, Khuri, Matthew, and Sinha

D. Class Periods

Lectures will be from 12:30 to 1:45 on Tuesday and Thursday.

It is my intention to follow the schedule printed elsewhere in the syllabus, but we may deviate slightly if it is in the best interest of the class.

E. Homework

Homework will be assigned and will account for 15% of your final grade. Since none of the books in this area are formal textbooks, homework will generally be working through examples and data sets. All homework will be late if handed in more than one week from the day assigned at the beginning of the class period, unless otherwise specified. There will be no credit for late homework (which includes homework handed in after the beginning of the class period). That being the case, if you will not be in class when a homework set is due, you should make arrangements to get it in prior to the last minute or have someone hand it in for you. Homework will need to be handed in as instructed by the TA. You may work in groups of two or three on homework assignments. If you choose to do so, please hand in only one copy of the assignment with all the names of those contributing on the assignment.

F. Exams

There will be 3 exams during the course. Exams will account for 75% of your grade (25% for each exam.) Exams handed in late will be graded with a maximum grade of 50%.

G. Simulation Project

You will work with a partner to produce a written project (3 or fewer pages of text, 4 or fewer with figures) that will consist of a simulation study over the following topic.

The LRT vs. the F-test

The project will account for 10% of your grade. Projects handed in late will be graded with a maximum grade of 50%.

The format for your project should be as follows. The headings would be

- i. Introduction
- ii. Data
- iii. Methods
- iv. Results
- v. Discussion

For this to be successful, each project should look at different covariance structures and/or different sample sizes and/or different percentages of missing data. Each project should examine both Type 1 error and power. We will explore this assignment in more detail when we get further into class.

H. Grades

1. Your final letter grade will be computed based on the following percentages:

94% and above	A	80–84%	B–	66–70%	D+
90–94%	A–	77–80%	C+	62–66%	D
87–90%	B+	74–77%	C	57–62%	D–
84–87%	B	70–74%	C–	57% and below	E

I. The Honor Code

While I do not believe that your eternal salvation is somehow associated with the length of your hair or your willingness to shave, I am absolutely convinced that your eternal well-being is closely associated with your willingness to be obedient and to keep covenants. With that in mind, I will take seriously your commitment to the Honor Code. I most fervently believe that temporal accomplishment is impacted positively by paying attention to eternal values, and intellectual growth of the richest kind can only be obtained by maintaining an eternal perspective.

Cheating is an activity from which no one benefits. My main criteria for inappropriate conduct in this regard is that you can certify that any material you hand in on an exam is your own work. I am happy for you to look at homework with your colleagues. Your simulation study will be completed with a partner. Using AI seems okay to me. If a problem can be solved only using AI then it is not a good problem and should be discarded as either a homework or exam problem. I will count on you to give me feedback on this.

J. Preventing and Responding to Sexual Misconduct

In accordance with Title IX of the Education Amendments of 1972, Brigham Young University prohibits unlawful sex discrimination against any participant in its education programs or activities. The university also prohibits sexual harassment — including sexual violence — committed by or against students, university employees, and visitors to campus. As outlined in university policy, sexual harassment, dating violence, domestic violence, sexual assault, and stalking are considered forms of "Sexual Misconduct" prohibited by the university.

University policy requires all university employees in a teaching, managerial, or supervisory role to report all incidents of Sexual Misconduct that come to their attention in any way, including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Incidents of Sexual Misconduct should be reported to the Title IX Coordinator at t9coordinator@byu.edu or (801) 422-8692. Reports may also

be submitted through EthicsPoint at <https://titleix.byu.edu/report> or 1-888-238-1062 (24-hours a day).

BYU offers confidential resources for those affected by Sexual Misconduct, including the university's Victim Advocate, as well as a number of non-confidential resources and services that may be helpful. Additional information about Title IX, the university's Sexual Misconduct Policy, reporting requirements, and resources can be found at <http://titleix.byu.edu> or by contacting the university's Title IX Coordinator.

K. Students with Disabilities

BYU is committed to providing reasonable accommodation to qualified persons with disabilities. If you have any disability that may adversely affect your success in this course, please contact the Services for Students with Disabilities Office at 422-2767. Services deemed appropriate will be coordinated with the student and instructor by that office.

L. Mental Health Services

Barriers to learning are created by stress, anxiety, family and relationship concerns, and personal crises. If stressful life events or mental health concerns are inhibiting your ability to participate in daily activities or leading to diminished academic performance, please contact the BYU Counseling and Psychological Services (CAPS) (1500 WSC, 801-422-3035, caps.byu.edu). CAPS provides individual, couples, and group counseling to students. These services are confidential and are provided by the university at no added cost to you. Professional psychologists and counselors who specialize in helping college students are available 24-hours a day to assist students in crisis; if you have an emergency during non-business hours (5 PM–8 AM), please contact BYU Police Dispatch (801-422-2222) who will put you in touch with a counselor.

Lecture Schedule

<u>Date</u>	<u>Topic</u>
9/3	Intro, Syllabus, AI expectations, Likelihood
9/8	Review Likelihood and the linear model, HW 1 maximum likelihood estimators
9/10	t and F tests from first principles, HW 2
9/15	Cell means model, missing cells, $WA^{-1}A\mu$
9/17	R (gls), the bond example, likelihood ratio test, HW 3
9/22	gls, compound symmetry, general cov
9/24	mixed model, X and Z
9/29	R (lmer), example in detail, lmerTest, HW 4
10/1	linear combinations of cell means, est. variances, pp
10/6	df, Satterthwaite, Kenward-Roger, SAS, F test, HW 5
10/8	REML Theory
10/13	Back to Bayes, HW 6
10/15	Nimble introduction, the bond example
10/20	repeated measures, freq and Bayesian
10/22	repeated measures, HW 7
10/27	In Class Exam
10/29	Random effects models, estimator for u
11/3	$\hat{V}(\hat{u})$, $\hat{V}(\hat{u} - u)$, HW 8
11/5	Random coefficients models, wheat data, R
11/5 — 11/10	First Take Home Exam
11/10	Choleski residuals
11/12	Choleski residuals, HW 10
11/17	Spatial data
11/19	Spatial data, HW 11
11/24	Simulation Study due
11/24	Smoothing, HW 12
12/1	Taping example
12/3	Hemodialysis example
12/3 — 12/8	Second Take Home Exam
12/8	Review
12/10	Devotional

The final will be from 11 am until 2 pm on Wednesday, December 16th.